SHUBHAN BHATIA

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EDUCATION

Ludwig-Maximilians-Universität München 2025 – Mar. 2027 (expected)
Master of Science in Astrophysics

New York University Abu Dhabi 2020 – 2024
Bachelor of Science in Physics
Founders' Day Award Recipient

PUBLICATIONS

S. Bhatia, L. Kimmig, R.-S. Remus, B. Seidel, K. Dolag, M. Kluge, S. Brough, **How Old Are You Really? Validating the ICL+BCG Fraction as a Dynamical Age Tracer of Galaxy Clusters in Projection**, in internal collaboration review; submission to A&A planned for November 2026.

M. Nori, **S. Bhatia**, A.V. Macciò, FUZZY-GASOLINE**: Cosmological hydrodynamical simulations of dwarf galaxy formation with Fuzzy Dark Matter**, arXiv:2411.09733; in preparation for resubmission.

RESEARCH EXPERIENCE

Graduate Research – Master's Thesis Mar. 2026 – Present
Research Supervisor: Apl. Prof. Dr. Klaus Dolag
University Observatory Munich (USM), LMU Munich

- Reconstructed hydrostatic mass profiles from synthetic X-ray observations of 42 local-universe zoom-in clusters. Found a small typical bias ($b \approx 0.05$–0.08) but a scatter of $\sigma_b \approx 0.10$ that grows with dynamical state, and a regression-based correction that halves the per-cluster mass error (12 to 6%).

Graduate Research – ICL+BCG Project Jul. 2025 – Present
Research Supervisors: Dr. Lucas Kimmig, Dr. Rhea-Silvia Remus and Apl. Prof. Dr. Klaus Dolag
University Observatory Munich (USM), LMU Munich

- Showed that the intracluster light and brightest cluster galaxy (ICL+BCG) fraction, a strong 3D tracer of cluster dynamical state, can be recovered in projection to ∼4.3 percentage points, with masking aperture, not projection angle, dominating the error budget.

Researcher Oct. 2024 – Nov. 2025
Research Supervisor: Prof. Joseph Gelfand
Center for Astrophysics and Space Science (CASS), New York University Abu Dhabi

- Built Swift/UVOT light curves for ∼750 ZTF-identified sources in IC 10 from a ∼5-year weekly monitoring campaign, identifying UV-detected, variable and periodic sources. Fitted their SEDs and found UV sources concentrated along the edges of HI holes.

Undergraduate Research – Bachelor's Thesis Sep. 2022 – Feb. 2025
Research Supervisors: Prof. Andrea Macciò and Dr. Matteo Nori
Galaxy Formation Group, CASS, New York University Abu Dhabi

- Produced a suite of 30 hydrodynamical and dark-matter-only zoom-in simulations of dwarf galaxies (10^9-$10^{11}\,M_\odot$) using the FUZZY-GASOLINE code on NYUAD's Jubail HPC cluster, part of the first high-resolution hydrodynamical galaxy-formation simulations with Fuzzy Dark Matter (FDM). Found delayed star formation in low-mass FDM halos relative to their Cold Dark Matter (CDM) counterparts from NIHAO, while other stellar observables remained broadly consistent.

Research Assistant May 2021 – Aug. 2021
Research Supervisor: Dr. Mohammad Ali-Dib
Center for Astro-Particle and Planetary Physics (CAP3), New York University Abu Dhabi

- Developed a supervised regression model to predict planetary orbital-stability limits in binary star systems, using training data from N-body simulations.

RESEARCH SOFTWARE

sim-highline: Galaxy Scaling Relations from Cosmological Simulations *(beta)*
App | Code | DOI

- Curated open database and web app of ∼1,600 published galaxy scaling relations from 20+ cosmological simulation suites (EAGLE, IllustrisTNG, SIMBA, COLIBRE and others), with full provenance for every relation and scripted checks against published values.

CONFERENCES, SCHOOLS AND PRESENTATIONS

Saas-Fee Advanced Course Jan. 2027
Saas-Fee, Switzerland – Registered participant: *From Particle Physics to Astrophysics: Cold Dark Matter and Beyond*

USM Science Day Oct. 2026
University Observatory Munich (USM), Munich, Germany – Talk: *Can Projected ICL+BCG Fractions Trace Cluster Formation Time?*

8th ICM Theory and Computation Workshop Jul. 2026
Brno, Czech Republic – Poster and flash talk: *Validating the ICL+BCG Fraction in Projection as a Dynamical Age Tracer of Galaxy Clusters Using the Lower Decks Zoom-in Simulations*

MSc Science Day, LMU Munich 2025–2026
Three research talks on intracluster light, galaxy-cluster assembly, and cosmological simulations.

TECHNICAL SKILLS

Programming: Julia, Python, C++, C, Fortran
Scientific computing: NumPy, SciPy, Matplotlib, Pandas, Astropy, PyTorch
High-Performance Computing: Parallel computing (OpenMP, MPI), Job scheduling (SLURM)
Simulation & Analysis tools: Gasoline2, OpenGADGET3, Pynbody, HEASoft, IRAF, ds9, Git, LaTeX
Languages: English (Native), Hindi (Fluent), German (Intermediate - B1), Arabic (Intermediate)

COMMUNITY ENGAGEMENT

Treasurer and Outreach Coordinator – *Space Bar* 2022 – 2024
Organized on- and off-campus astronomy outreach events, including an observatory visit, and secured recurring funding for *Space Bar*, NYUAD's student astronomy interest group.

Podcast Co-host – *Spaced Out* 2023 – 2024
Co-hosted and coordinated Season 3, handling guest selection and episode preparation for a public astronomy podcast.

REFERENCES

Apl. Prof. Dr. Klaus Dolag	USM, LMU München	dolag@usm.lmu.de
Dr. Rhea-Silvia Remus	USM, LMU München	rhea@usm.lmu.de
Prof. Andrea Macciò	NYU Abu Dhabi	maccio@nyu.edu
Dr. Lucas Kimmig	University of Nottingham	lucas.kimmig@nottingham.ac.uk